

ELEC5305 Project Feedback Two

Ryan Hu | SID 550275044 | Attention-Based Temporal Localisation of Musical Instruments

GitHub Project Site: <https://decarre.github.io/elec5305-project-550275044/>

Brief project description

This project investigates whether a weakly supervised multi-label audio model can identify instruments in polyphonic music and recover when each instrument is active. A shared frame encoder is used for a controlled comparison of global mean pooling and instrument-specific attention pooling. Models receive clip-level instrument labels during training; time-aligned MedleyDB activation confidence is reserved for temporal evaluation.

Achievements to date

- **Implementation:** log-mel feature extraction; matched mean, max and AttentionMIC-style models; separate frame probabilities, attention weights and clip predictions; temporal post-processing; and a validate/train/evaluate command-line workflow.
- **Leakage prevention:** deterministic class-aware artist splits, stable track/clip manifests created after splitting, validation-only threshold selection, checkpoint reload, and frozen-threshold test evaluation.
- **Dataset audit:** official MedleyDB v1/v2 metadata contains 196 tracks from 116 artists; 117 tracks have matching v2 activation-confidence files. Both public sample mixes and all eight expected instrument stems passed audio/annotation alignment checks.
- **Verification:** 13 automated tests pass, covering model aggregation semantics, metrics, artist-disjoint manifests, checkpoint behaviour and temporal processing.

Preliminary engineering result

Model	Initial BCE	Final BCE	Micro-F1	Macro-F1
Mean pooling	0.4715	0.0357	0.8721	0.7717
Attention pooling	0.4509	0.0135	0.9977	0.9967

These values are training-set diagnostics from 230 non-overlapping two-second clips from two public sample songs. Training and measurement used the same clips. They verify the two model paths end to end and do not estimate generalisation or establish model superiority.

Current limitation and next step

Access to the full MedleyDB audio has been requested and is awaiting approval. After access is granted, complete artists will be assigned to train, validation and test before clipping. Thresholds and temporal post-processing will be selected only on validation artists, then clip-level and frame-level metrics will be reported once on held-out artists.

Project resources

Repository: <https://github.com/Decarre/elec5305-project-550275044>

Project Site: <https://decarre.github.io/elec5305-project-550275044/>